

Bohdan Hirianskyi

Machine Learning · Python · PhD in Computer Science

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SUMMARY

In August 2026 I defended a PhD in Computer Science at Igor Sikorsky Kyiv Polytechnic Institute. The dissertation studies how evolutionary algorithms optimize neural networks: four methods that prune and compress networks, tune hyperparameters, and merge models. Author of 5 research publications, including a paper in a Scopus-indexed journal. From September 2025 to June 2026 I worked as a junior software developer. I taught a university course on algorithm design. I am looking for applied, product-focused ML work.

SKILLS

Programming: Python (primary); C / C++ (basic)

Machine Learning & Data Science: PyTorch, scikit-learn, NumPy, SciPy, pandas, Matplotlib; evolutionary computation (CMA-ES, DEAP, sklearn-genetic); neural-network pruning and compression, hyperparameter optimization (Optuna), NAS, AutoML; statistical hypothesis testing (Friedman, Wilcoxon, Nemenyi tests).

Tools & Infrastructure: Git, Docker (basic), Google Colab, Claude Code.

DISSERTATION

PhD dissertation (122 Computer Science): "Optimization of Parameters and Structure of Neural Networks Using Evolutionary Algorithms". Defended on 28 August 2026 before one-time specialized academic council DF 26.002.419 at Igor Sikorsky KPI; unanimous vote.

TESA-26. Weight pruning. An evolutionary search (CMA-ES) decides what share of weights each layer keeps. Best Friedman rank (2.04) among 5 methods on 6 datasets; significantly better than SparseGPT, WANDA and RIA.

GFCS. Converts a sparse network into a compact dense one without training data or retraining. Compresses the model 7.6× and preserves quality on all 8 datasets studied.

SACMA-DAC and SACMA-MAB. Hyperparameter search. CMA-ES explores candidates on a surrogate model (Random Forest), and only one candidate per iteration gets a full training run. Ranked 1st and 2nd among 9 methods on 43 tasks, with the fastest convergence.

ENT. A genetic algorithm merges two networks trained on different classes into a single model. On MNIST the merged model recognizes all 10 classes; accuracy after calibration is 0.917.

Applicability limit. Showed that evolutionary algorithms work well for low-dimensional structural decisions, while gradient methods remain better at training network weights.

Deployment. TESA-26, GFCS and SACMA are used at SD Profzviazok LLC (deployment certificate, 5 May 2026).

Code and experiment results

PUBLICATIONS

1 Scopus-indexed paper, 3 papers in Ukrainian peer-reviewed journals, 1 conference abstract.

Effectiveness of evolutionary algorithms in neural-network optimization (2026)

Compared evolutionary (CMA-ES), gradient-based and Bayesian methods on unconstrained minimization, structured pruning and hyperparameter optimization. CMA-ES + SynFlow with energy compensation gave the best quality under extreme pruning (98%, F1 = 0.81). Reported the limits honestly: Adam trains weights better, and magnitude pruning wins at moderate sparsity.

Hirianskyi, B., & Bulakh, B. (2026). Effectiveness of evolutionary algorithms in neural network optimization tasks: Unconditional minimization, pruning, and hyperparameter optimization. Control, Navigation and Communication Systems, 1(83), 45–50. <https://doi.org/10.26906/sunz.2026.1.045>

Neuroevolutionary synthesis of neural networks (2025, in Ukrainian)

Implemented two neuroevolution algorithms from scratch, memetic and hybrid, that automatically select network architecture and weights. On non-linear binary-classification tasks (XOR, Moons, Breast Cancer) they outperformed a fixed-architecture perceptron. Python, NumPy, scikit-learn.

Hirianskyi, B., & Bulakh, B. (2025). [Creating neural networks through neuroevolution and automated learning using evolutionary algorithms]. Science and Technology Today, 8(49), 1214–1227. [https://doi.org/10.52058/2786-6025-2025-8\(49\)-1214-1227](https://doi.org/10.52058/2786-6025-2025-8(49)-1214-1227)

Evolutionary optimization of neural networks (2024)

Applied genetic algorithms (DEAP, sklearn-genetic) to optimize the architecture, hyperparameters and weights of a multilayer perceptron on the Wine Quality dataset. Reported honestly that the evolutionary approach did not, on average, beat the baseline in quality and required far more computation.

Hirianskyi, B., & Bulakh, B. (2024). Using evolutionary algorithms for training and optimizing neural networks. Science and Technology Today, 8(36), 808–823. [https://doi.org/10.52058/2786-6025-2024-8\(36\)-808-823](https://doi.org/10.52058/2786-6025-2024-8(36)-808-823)

A review of evolutionary algorithms for neural-network synthesis and training (2023, Scopus-indexed)

Systematized how genetic algorithms are applied to neural networks: feature selection, weight optimization, changes to connection structure, and optimizer tuning.

Hirianskyi, B., & Bulakh, B. (2023). A review of practice of using evolutionary algorithms for neural network synthesis and training. Technology Audit and Production Reserves, 4(2(72)), 22–26. <https://doi.org/10.15587/2706-5448.2023.286278>

International conference abstract (Athens, Greece, 24–26 December 2025, in Ukrainian)

Hirianskyi, B. (2025). [Using evolutionary algorithms for neural network synthesis and training: State of the art and prospects]. Science and Information Technologies in the Modern World, 165–168. <https://doi.org/10.70286/isu-24.12.2025.006>

OPEN SOURCE

The applications were built together with the Claude Code agent.

Bees Are Not About Honey

Popular-science book about bees told through memes: 48 memes, 13 chapters, in Ukrainian. Website with PDF and EPUB editions.

HTML, CSS, JavaScript · [Code](#) [Open](#)

DupScan

Duplicate file and folder finder for macOS.

Python, PySide6 · [Code](#)

ApexCam

Camera app for Android.

Kotlin, Compose, Camera2 and CameraX · [Code](#)

Kotyk za kermom

Browser driving game set on the streets of Obolon, Kyiv.

JavaScript, Vite · [Code](#) [Open](#)

NO STING

Bilingual web comic about bees.

HTML, CSS, JavaScript · [Code](#) [Open](#)

Android Mover

File transfer between Android and Mac over adb.

Swift, SwiftUI · [Code](#)

ShouldIPost

Predicts how a video will perform before it is posted.

Python, scikit-learn, FastAPI · [Code](#)

EXPERIENCE

Junior Software Developer

KILOBYTE1024 LLC (ТОВ «КИЛОБАЙТ1024»), Kyiv · 09.2025 – 06.2026

Teaching Assistant

Igor Sikorsky Kyiv Polytechnic Institute · during PhD studies

Led practical and laboratory sessions for the course “Design and Analysis of Computational Algorithms”: complexity analysis, data structures, algorithm-design strategies.

Systems Administrator

Department of System Design, Igor Sikorsky Kyiv Polytechnic Institute · 11.2017 – 12.2018

Administered 20+ servers (Windows Server) and 70+ workstations: virtualization (VMware, VirtualBox), Active Directory, networking and monitoring.

EDUCATION

Igor Sikorsky Kyiv Polytechnic Institute · Institute for Applied System Analysis (IASA) · Department of System Design

PhD (Doctor of Philosophy), Computer Science (122)

2022 – 2026

Dissertation defended on 28 August 2026 (council DF 26.002.419). Supervisor: Assoc. Prof. B. Bulakh, PhD

MSc, Computer Science

2020 – 2021

Diploma with honours. Thesis: “Estimating patient age as a health measure from ECG” — machine / deep learning

BSc, Computer Science

2016 – 2020

Thesis: “Using machine learning to detect cardiovascular diseases”

LANGUAGES

Ukrainian — native English — B2